# The Operational Amplifier

**EEE103 ELECTRICAL CIRCUITS (Part 2)**
**Week 5**
**S1, 2026**

Lecturer: Xuchen Wang
Xuchen.Wang@xjtlu.edu.cn



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# Overview of Part 2

- Part 2 introduces operational amplifier and discusses circuit impedance components and dc circuit transients.

  Week 5: The operational amplifier

  Week 6: Capacitor and Inductor

  Week 7: Basic RC and RL circuit

  Understanding!!!

- In Class Assignment: Week 8, Thursday, 18:00 - 20:00.
- Office hour: Wednesday, 13:00-17:00 (Location: EE312)



# Content

1. The characteristics of operational amplifier
2. The inverting amplifier
3. Other basic operational amplifier circuits:
    - The non-inverting amplifier
    - The voltage follower
    - The summing amplifier
    - Cascaded operational amplifier circuit
4. Comparators and the Schmitt triggers



# Content

**1. The characteristics of operational amplifier**

2. The inverting amplifier

3. Other basic operational amplifier circuits:

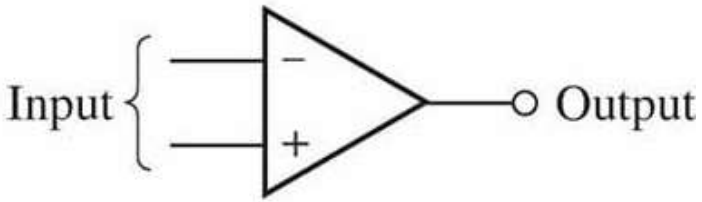
- The non-inverting amplifier
- The voltage follower
- The summing amplifier
- Cascaded operational amplifier circuit

4. Comparators and the Schmitt triggers



# Introduction

- An operational amplifier (op amp) is an integrated circuit that amplifies the difference between input voltages and produces a single output.



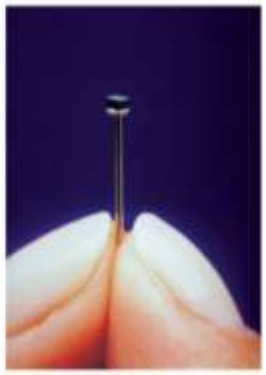

- The origin of operational amplifier is to perform mathematical operations such as addition, subtraction, multiplication, division, differentiation, and integration for analog computers. This is where the term "operational amplifier" comes from!
- The op amp finds daily usage in a large variety of electronic applications. Modern op amps are constructed using transistors in conjunction with resistors and capacitors to perform the desired characteristics.




A vacuum tube op amp



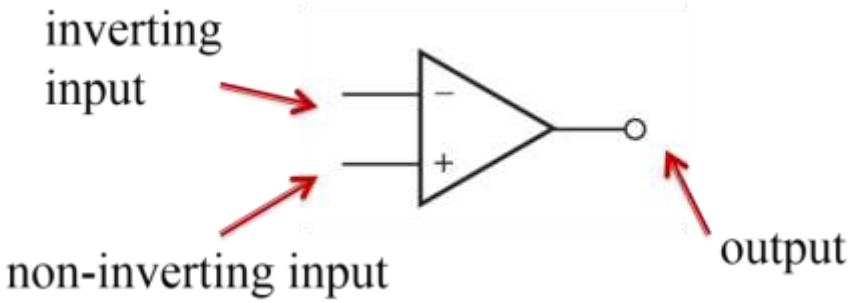
A LMV321 op amp



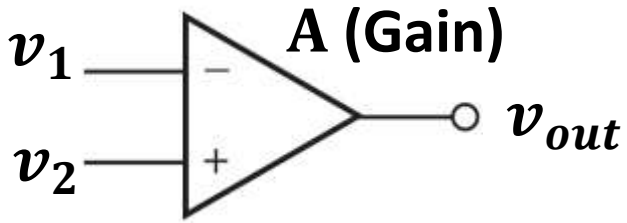
A LMC6035 op amp

# Introduction

- The symbol of op amps have three principal terminals (two inputs and one output):



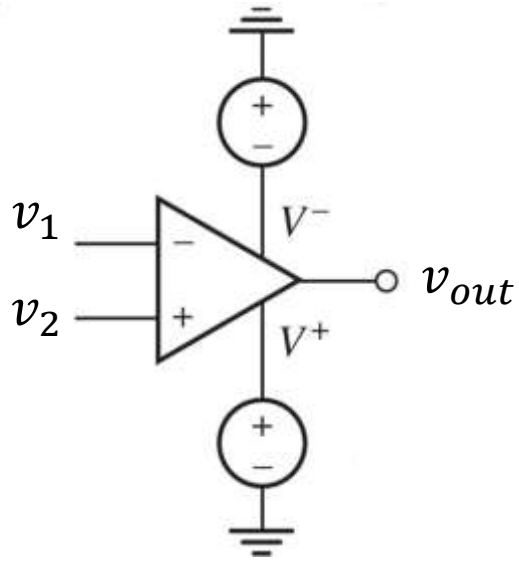

- The op amp senses the difference between two input signals and amplifies this difference to produce an output signal.

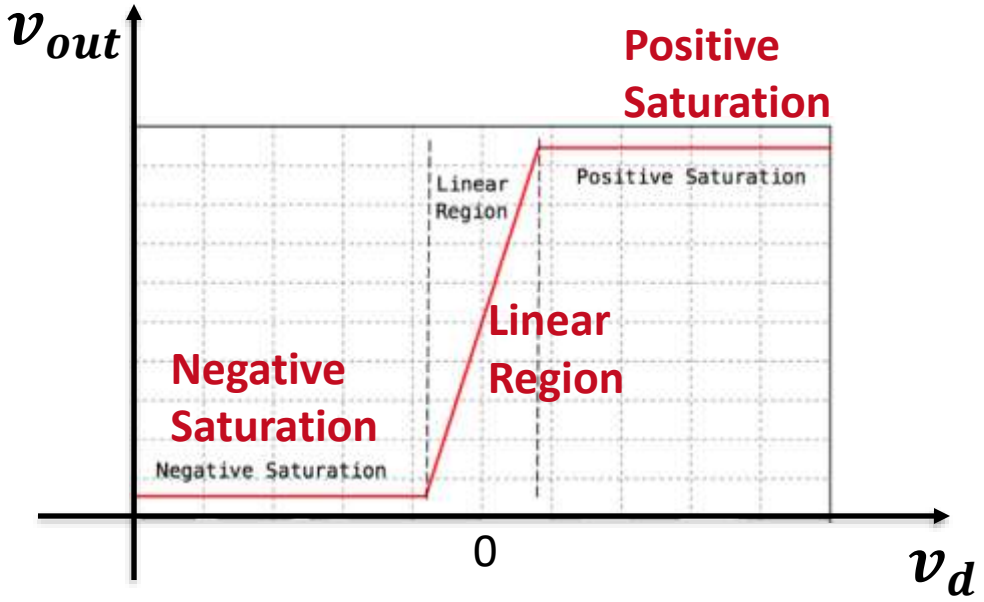



$$v_{out} = Av_d = A(v_2 - v_1)$$

# Power supply and saturation

- An op amp requires power supplies.
- Usually, equal and opposite voltages are connect to V+ and V−
- The output of op amp is a linear response bounded by the positive and negative saturation regions




- Typical values: 5 to 24 volts.
- Power supply ground must be the same as the signal ground

# Content

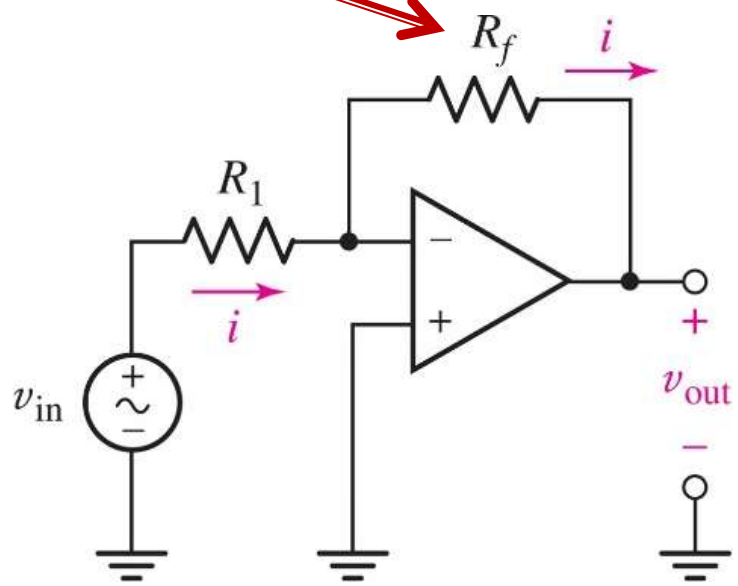
1. The characteristics of operational amplifier
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# Feedback amplifier

- The enormous but unpredictable gain of the op amp is made usable through negative feedback.

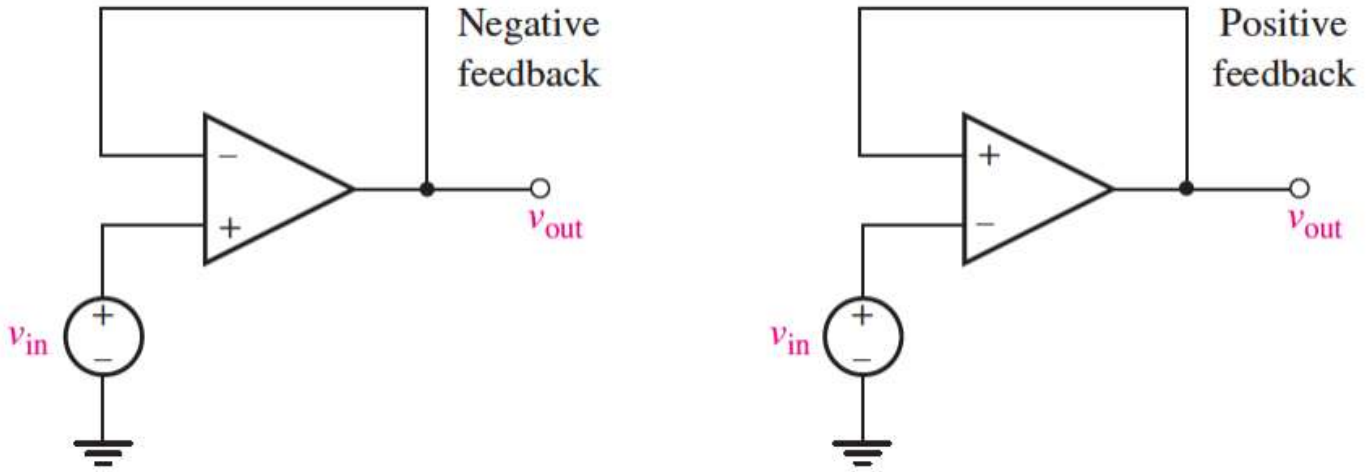
this “feedback” resistor
allows the output to affect
the input terminal.




Inverting amplifier

# Feedback amplifier

- Circuits with direct electrical connection between output and input are called closed-loop. Both of the negative feedback and positive feedback configurations are closed-loop configurations.




- Negative feedback circuits are usually used in operational amplifier circuits.

**Why?**

# Negative Feedback amplifier

- In case of negative feedback, the output voltage is:

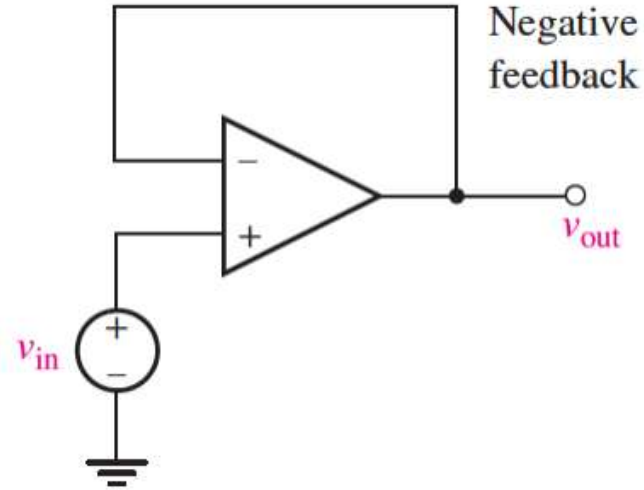
$$v_{out} = Av_d = A(v_{in} - v_{out}) \qquad [1]$$

  - If $v_{out} < v_{in}, v_d > 0$, $v_{out}\uparrow$, $|v_{in} - v_{out}|\downarrow$
  - If $v_{out} > v_{in}, v_d < 0$, $v_{out}\downarrow$, $|v_{in} - v_{out}|\downarrow$

Negative feedback decreases the input and can lead to a stable state !

- Rearranging Eq(1), we obtain $V_{out}$

$$v_{out} = \left(\frac{A}{A+1}\right) v_{in} \qquad [2]$$




where we see that the output will be slightly smaller than the input, then fed back into the inverting terminal of the op amp. In limit of large A, the output will stabilize to $v_{out} \approx v_{in}$.

# Positive Feedback amplifier

- Recalling A is the very large voltage gain of the op amp, (assumed infinite for ideal case), the output voltage is:

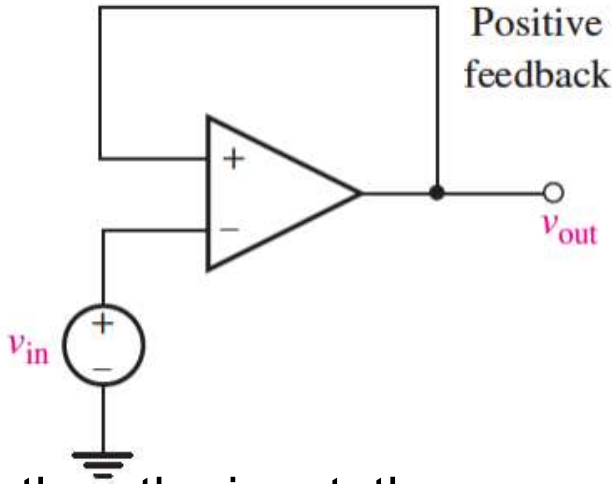
$$v_{out} = A v_d = A(v_{out} - v_{in}) \qquad [1]$$

  - If $v_{out} < v_{in}, v_d < 0, v_{out} \downarrow, |v_{out} - v_{in}| \uparrow$
  - If $v_{out} > v_{in}, v_d > 0, v_{out} \uparrow, |v_{out} - v_{in}| \uparrow$

Positive feedback increases the input and can lead to instability / op amp saturation !

- Rearranging Eq(1), We obtain $V_{out}$

$$v_{out} = \left(\frac{A}{A-1}\right) v_{in} \qquad [2]$$




where we see that the output will be slightly larger than the input, then fed back into the non-inverting terminal of the op amp.

# Content

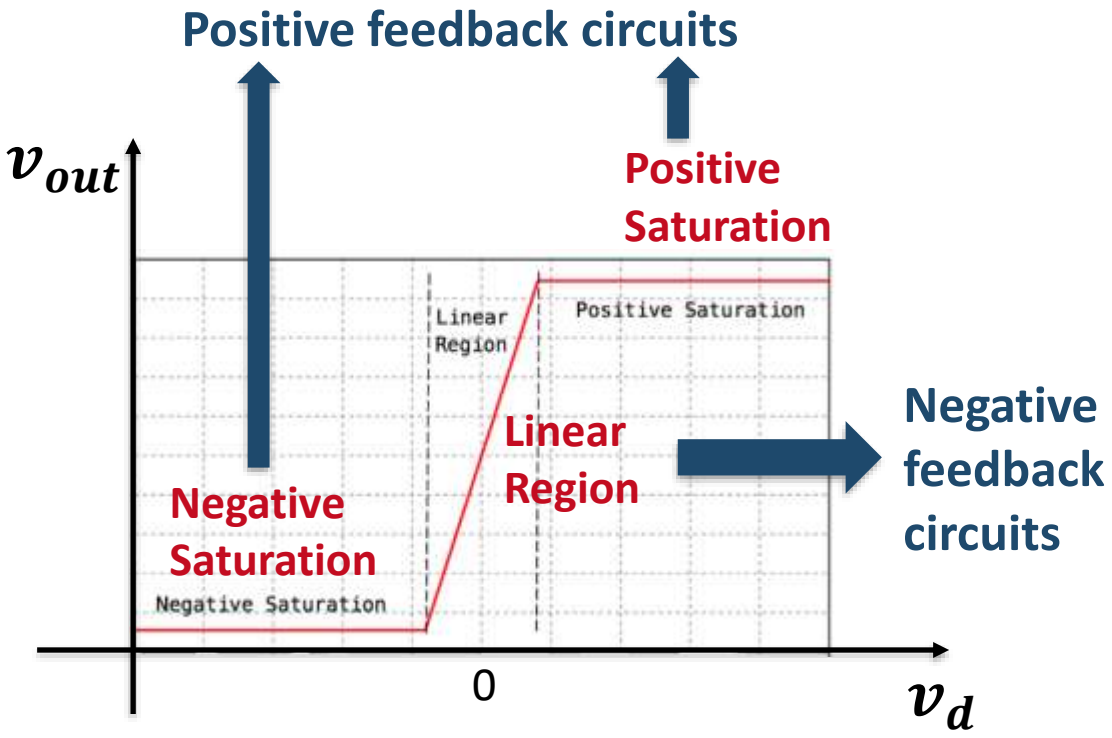
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Negative feedback circuits

Positive feedback circuits

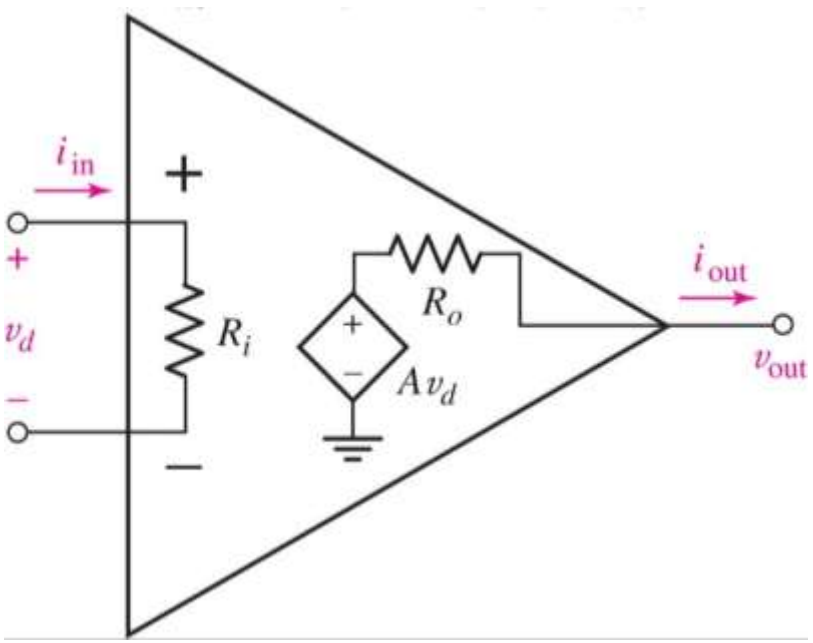


# Feedback circuit and saturation



# The Ideal Op Amp

- The internal differential gain $A$ is considered to be infinite.
- The differential input voltage $v_d$ is assumed to be zero. If $A$ is very large and if the output $v_{out}$ is finite, then the two input voltages must be nearly equal.
- The effective input resistance to the op-amp is assumed to be infinite, so the input current, $i_{in}$ , is essentially zero.




- The output resistance $R_o$ is assumed to be zero in the ideal case, so the output voltage is connected directly to the dependent voltage source, and the output voltage is independent of any load connected to the output.

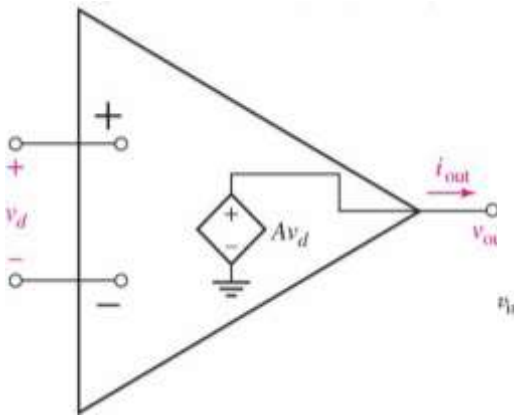
In summary: $A = \infty$, $R_o = 0\ \Omega$, and $R_i = \infty\ \Omega$, the op amp behaves according to the ideal op amp rules. ($v_d = 0$ *and* $i_{in} = 0$)

# Two Ideal Op Amp Rules

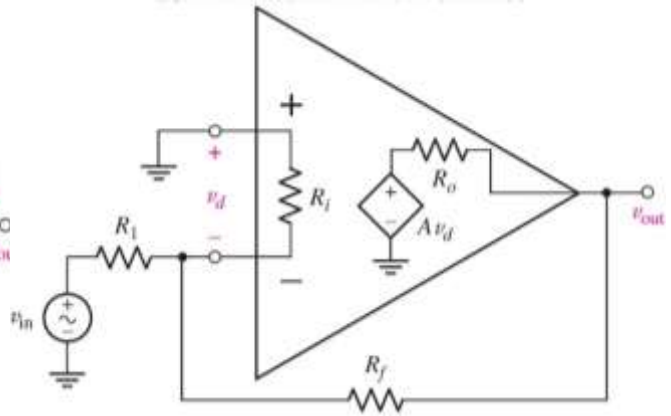
- **Rule 1:** No current ever flows into either input terminal. (Current can flow at output terminal!)
- **Rule 2:** There is no voltage difference between the two input terminals.

$$v_{out} = Av_d \neq 0$$

Note the two rules here are for the op amp with external circuit analysis!




A ideal Op Amp model




Negative feedback amplifier

The Op amp acts to make this happen!

# The Inverting Amplifier

Apply KVL and KCL, Ohm's law, and the ideal op amp rules to find $v_{out}$

- According to op amp rule 1, apply KVL:

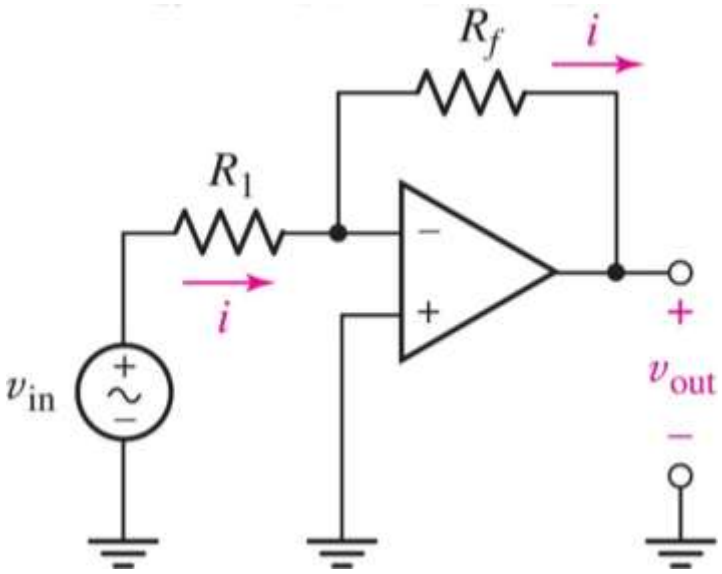
$$-v_{in} + R_1 i + R_f i + v_{out} = 0 \qquad [1]$$

- According to op amp rule 2, apply KCL:

$$i = \frac{v_{in} - 0}{R_1} \qquad [2]$$

- Combing Eq.[1] and [2], we obtain $v_{out}$:

$$v_{out} = -\frac{R_f}{R_1} v_{in} \qquad [3]$$

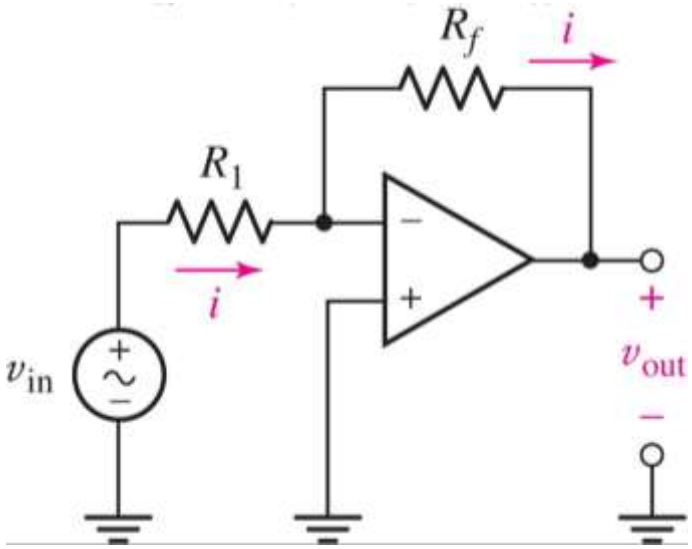



The resulting answer shows that the output $v_{out}$ is proportional to the input $v_{in}$ by the factor of $-\frac{R_f}{R_1}$, or amplifies the input by a negative constant defined by resistor values.

# Example 1

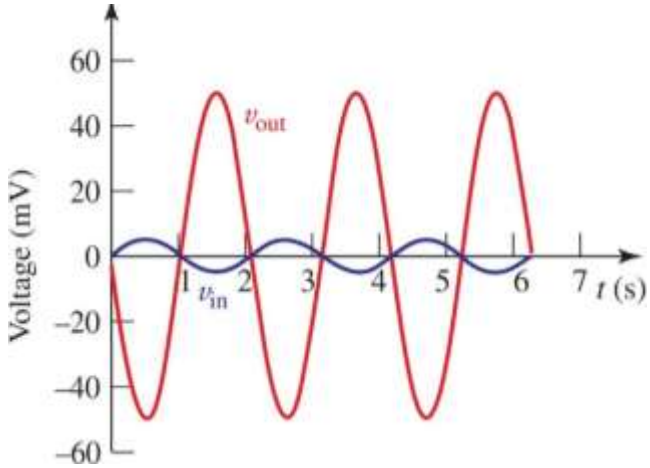
Derive an expression for inverting amplifier $v_{out}$ in terms of $v_{in}$ the circuit show below.

$$v_{in}(t) = 5sin3t\ mV, R_f = 47k\Omega, R_1 = 4.7k\Omega$$




Solution:

$$v_{out}(t) = -\frac{R_f}{R_1}v_{in} = -50sin3t\ mV$$

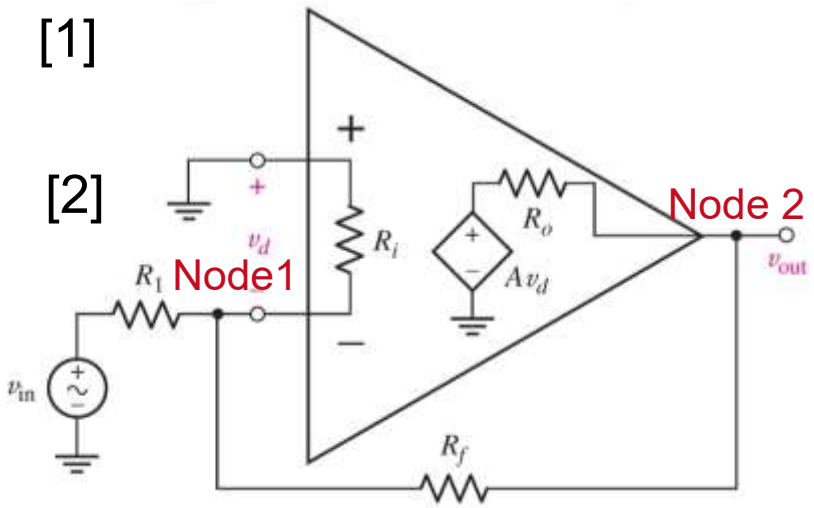


# Inverting Amplifier with a Real Op Amp

- Note we are no longer using ideal op amp rules for real op amp model, thus we write two nodal equations:

At node 1: $\frac{-v_d - v_{in}}{R_1} + \frac{-v_d - v_{out}}{R_f} + \frac{-v_d}{R_i} = 0$ [1]

At node 2: $\frac{v_{out} + v_d}{R_f} + \frac{v_{out} - Av_d}{R_0} = 0$ [2]




- According to Eq.[1] and [2], and eliminate $v_d$, we obtain $v_{out}$:

$$v_{out} = \left[\frac{R_0 + R_f}{R_0 - AR_f}\left(\frac{1}{R_1} + \frac{1}{R_f} + \frac{1}{R_i}\right) - \frac{1}{R_f}\right]^{-1} \frac{v_{in}}{R_1} \quad [3]$$

- Now if we allow $A = \infty$, $R_o = 0\ \Omega$, and $R_i = \infty\ \Omega$, Eq.[3] reduces to

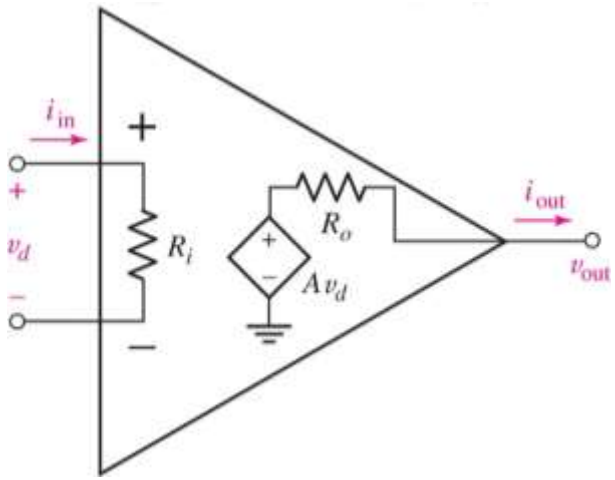
$$v_{out} = -\frac{R_f}{R_1} v_{in} \quad [4]$$

which is exactly what we derived for the inverting amplifier when the op amp is ideal.

# A more detailed Op Amp model

The practical characteristics of op amp:

- The open loop gain $A$ is the range of $10^5$ to $10^6$.
- The effective input resistance $R_i$ is the range of MΩ to TΩ.
- The output resistance $R_o$ is in the range of several to dozens ohms.




| Part Number | μA741 | LM324 | LT1001 | LF411 | AD549K |
|---|---|---|---|---|---|
| Description | General purpose | Low-power quad | Precision | Low-offset, low-drift JFET input | Ultralow input bias current |
| Open-loop gain *A* | $2 \times 10^5$ V/V | $10^5$ V/V | $8 \times 10^5$ V/V | $2 \times 10^5$ V/V | $10^6$ V/V |
| Input resistance | 2 MΩ | * | 100 MΩ | 1 TΩ | 10 TΩ |
| Output resistance | 75 Ω | * | * | ~1 Ω | ~15 Ω |
| Input bias current | 80 nA | 45 nA | 0.5 nA | 50 pA | 75 fA |
| Input offset voltage | 1.0 mV | 2.0 mV | 7 μV | 0.8 mV | 0.150 mV |
| CMRR | 90 dB | 85 dB | 110 dB | 100 dB | 100 dB |

* Not provided by manufacturer.

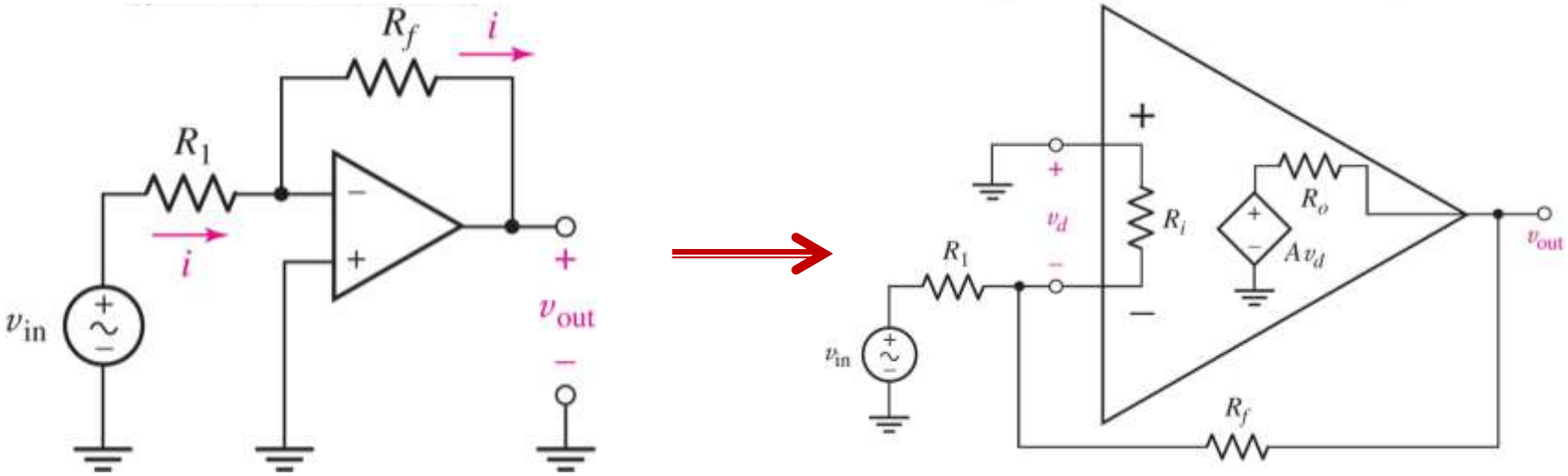
A more detailed Op Amp Model

# Example 2

Using the following values for *u*A741 op amp, reanalyze the inverting amplifier circuit and find $v_{out}$.

$$v_{in}(t) = 5sin3t\ mV, R_f = 47k\Omega, R_1 = 4.7k\Omega$$

For a 741 op amp: $A = 2 \times 10^5$, $R_i = 2\text{M}\Omega$, $R_o = 75\Omega$




Solution: $v_{out}(t) = -49.9977sin3t\ mV$

(Ideal op amp: $v_{out}(t) = -50sin3t\ mV$)

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# The Non-inverting Amplifier

Solve $V_{out}$, use KVL, KCL, and op amp rules.

- According to op amp rule 1:

  at node *a*: $0 = \frac{v_a}{R_1} + \frac{v_a - v_{out}}{R_f}$ [1]

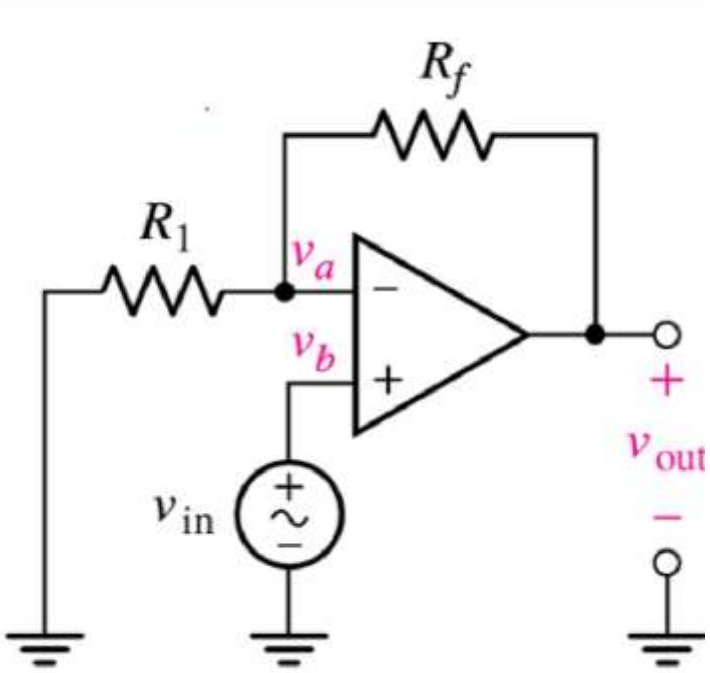
  at node *b*: $v_b = v_{in}$ [2]

- According to op amp rule 2, since $v_a = v_b = v_{in}$,

  $$0 = \frac{v_{in}}{R_1} + \frac{v_{in} - v_{out}}{R_f} \quad [3]$$

- Rearranging Eq.[3], We obtain $v_{out}$

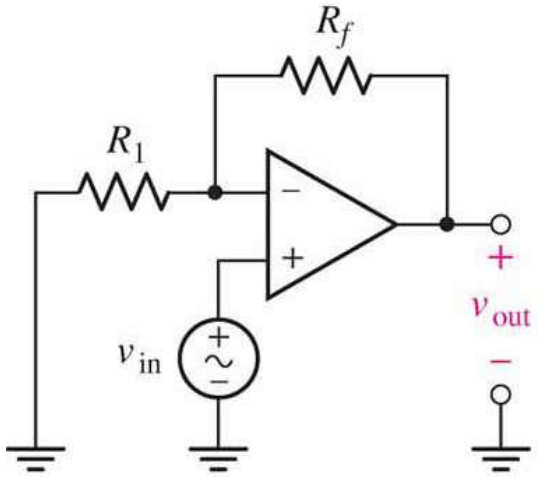
  $$v_{out} = (1 + \frac{R_f}{R_1}) v_{in} \quad [4]$$



# Example 3

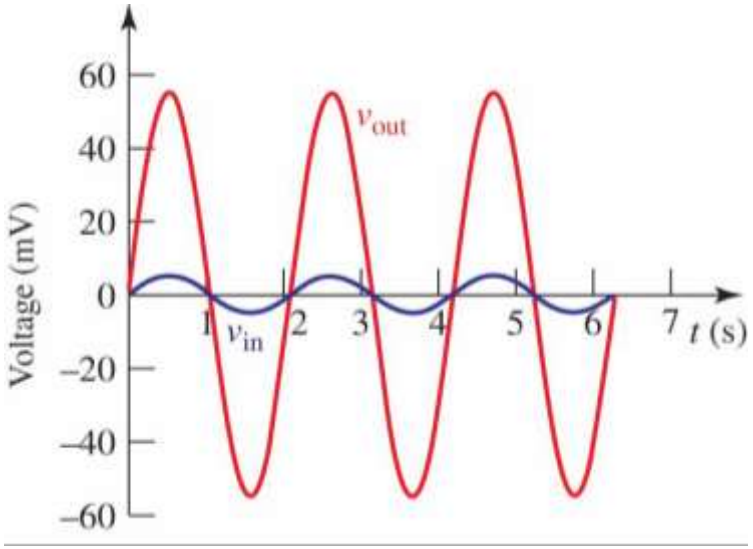
Derive an expression for non-inverting amplifier $V_{out}$ in terms of $V_{in}$ the circuit show below.

$$v_{in}(t) = 5sin3t\ mV, R_f = 47k\Omega, R_1 = 4.7k\Omega$$



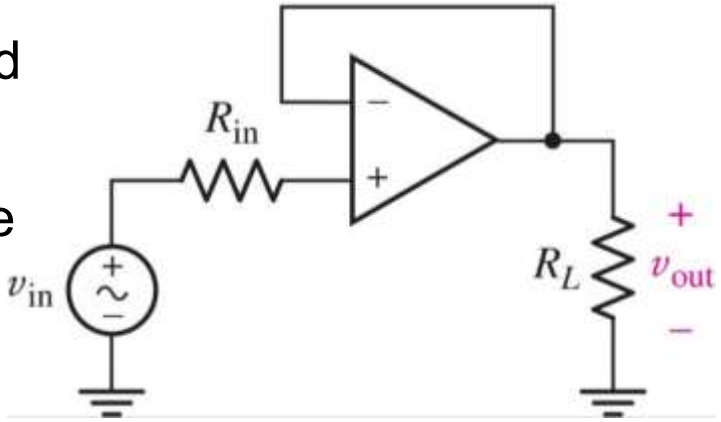

Solution:

$$v_{out}(t) = (1 + \frac{R_f}{R_1})v_{in} = 55sin3t\ mV$$



# The Voltage Follower

- The voltage follower is a noninverting amplifier
- $R_1$ set to $\infty$ and $R_f$ set to zero.
- Output is identical to input in both sign and magnitude.
- Voltage follower draws no current from the input.
- Act as buffer between $v_{in}$ and $v_{out}$




**What is the output voltage?**

Solution: $v_{out} = v_{in}$

# The Summing amplifier

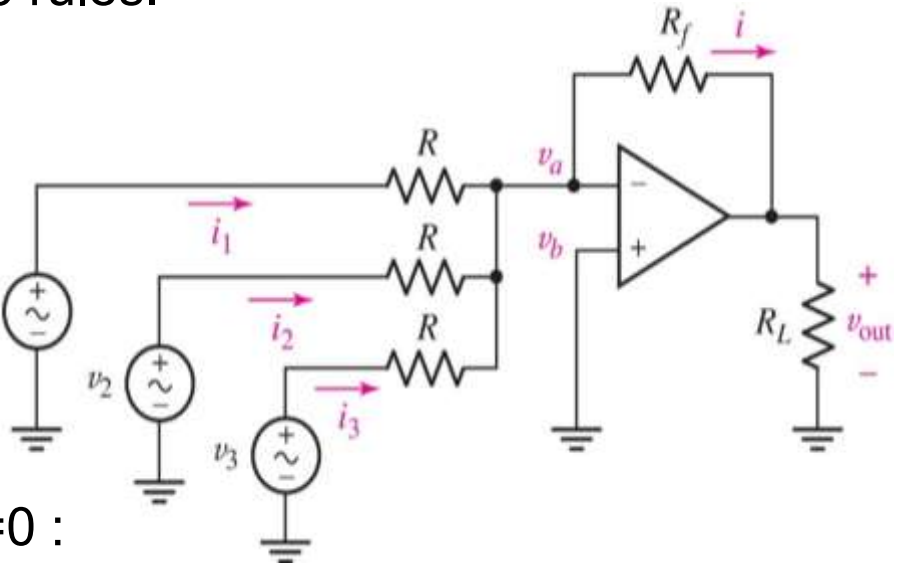
Solve $V_{out}$, use KVL, KCL, and op amp rules.

- According to op amp rule 1:

$$i_1 + i_2 + i_3 = i \qquad [1]$$

- Therefore, at the node labeled $V_a$:

$$\frac{v_1 - v_a}{R} + \frac{v_2 - v_a}{R} + \frac{v_3 - v_a}{R} = \frac{v_a - v_{out}}{R_f} \qquad [2]$$




- According to op amp rule 2, $V_a = V_b = 0$ :

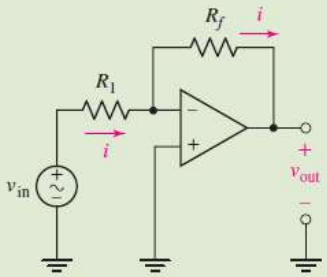
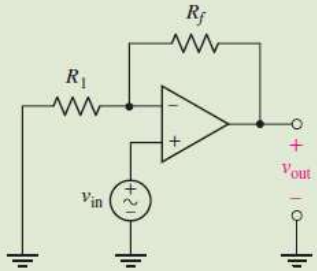
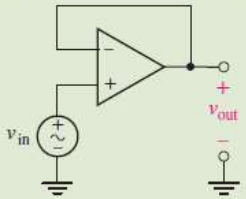
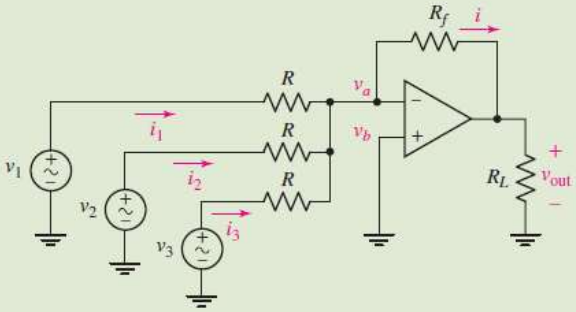
$$\frac{v_1}{R} + \frac{v_2}{R} + \frac{v_3}{R} = -\frac{v_{out}}{R_f} \qquad [3]$$

- Rearranging Eq.[3], We obtain $V_{out}$

$$v_{out} = -\frac{R_f}{R}(v_1 + v_2 + v_3) \qquad [4]$$

This amplifier performs the operation of adding. It also introduces a gain of $-\frac{R_f}{R}$ (same with inverting amplifier).

# Summary of basic Op Amp Circuits

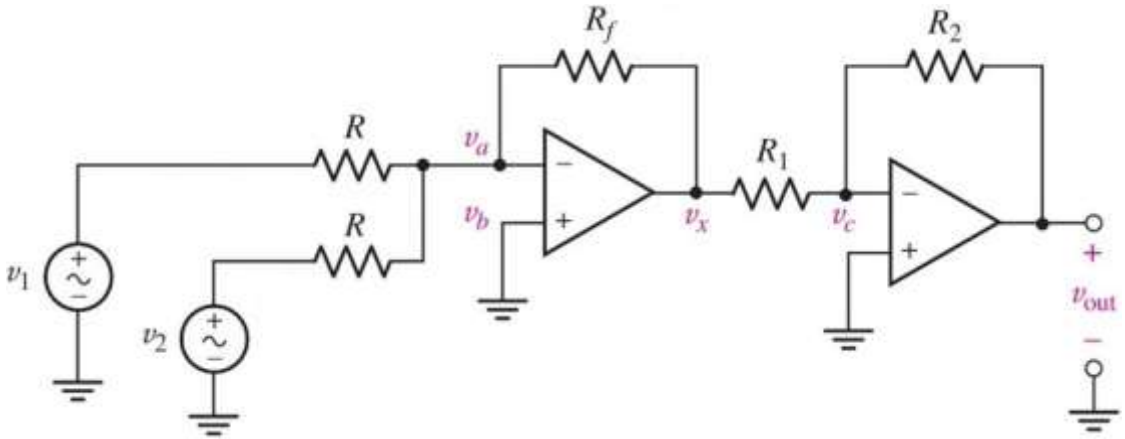
| Name | Circuit Schematic | Input-Output Relation |
|---|---|---|
| Inverting Amplifier |  | $v_{out} = -\frac{R_f}{R_1} v_{in}$ |
| Noninverting Amplifier |  | $v_{out} = \left(1 + \frac{R_f}{R_1}\right) v_{in}$ |
| Voltage Follower (also known as a Unity Gain Amplifier) |  | $v_{out} = v_{in}$ |
| Summing Amplifier |  | $v_{out} = -\frac{R_f}{R}(v_1 + v_2 + v_3)$ |

# Cascaded stages

Solve $V_{out}$, use KVL, KCL, and op amp rules.

- According to op amp rule 1, at the node labeled $v_c$

$$\frac{v_c - v_x}{R_1} + \frac{v_c - v_{out}}{R_2} = 0 \qquad [1]$$




- According to op amp rule 2, $v_c = 0$:

$$\frac{v_x}{R_1} + \frac{v_{out}}{R_2} = 0 \qquad [2]$$

- According to op amp rule 1, at the node labeled $v_a$:

$$\frac{v_a - v_x}{R_f} + \frac{v_a - v_1}{R} + \frac{v_a - v_2}{R} = 0 \qquad [3]$$

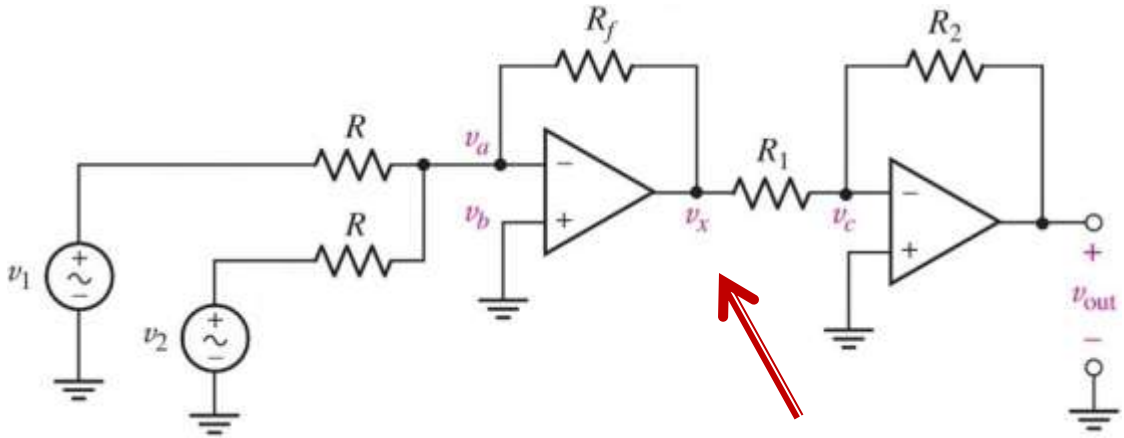
- According to op amp rule 2, $v_a = 0$, Eq.[3] becomes:

$$\frac{v_x}{R_f} + \frac{v_1}{R} + \frac{v_2}{R} = 0 \qquad [4]$$

# Cascaded stages

- Combing Eq.[2] and [4], We obtain $v_{out}$

$$v_{out} = \frac{R_2}{R_1}\frac{R_f}{R}(v_1 + v_2) \qquad [5]$$




This voltage is not affected by the circuit on the right.

- Op amps can be combined in stages to create the desired relationship between the outputs and the inputs .
- The cascaded circuit act as a summing amplifier, but without inverting the sign between the input and output.

# Content

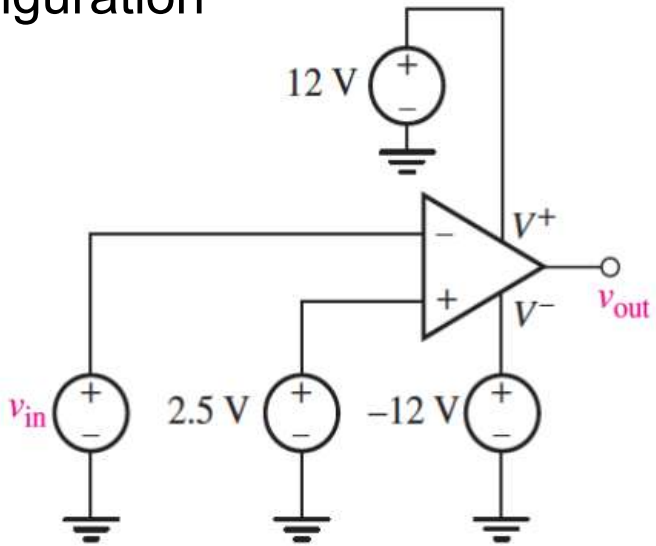
1. The characteristics of operational amplifier
2. The inverting amplifier
3. Other basic operational amplifier circuits:
    - The non-inverting amplifier
    - The voltage follower
    - The summing amplifier
    - Cascaded operational amplifier circuit

**4. Comparators and the Schmitt triggers**



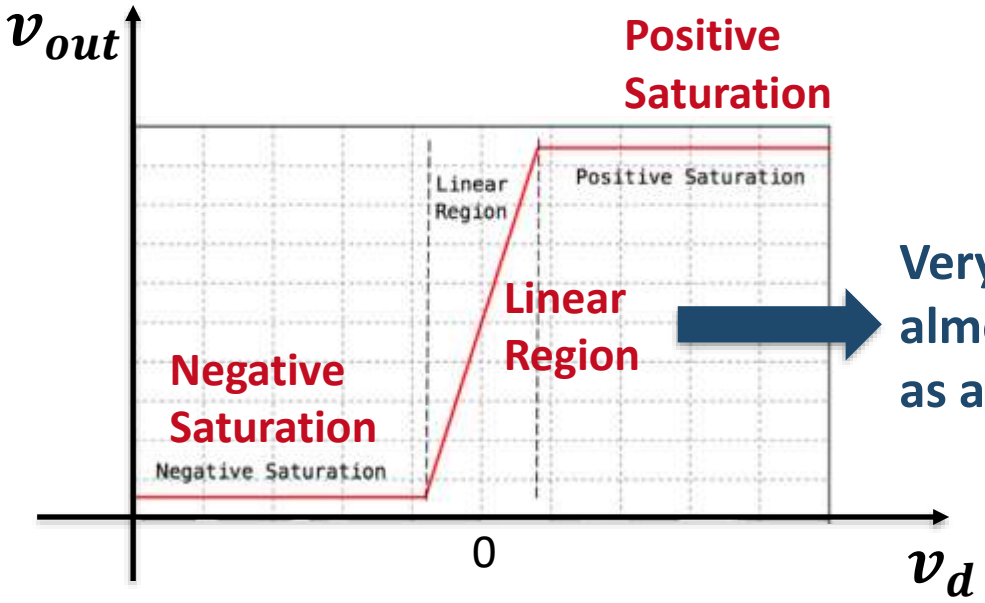
# Comparator

- Closed loop is the preferred method of using op amp as an op amp (all the previous op amp circuits)
- Op-amp can also be used an open-loop configuration with different design: comparators!
- Since the op amp has a very large open-loop gain A, it can force the op amp into saturation with small difference.
- Thus, the output depends on how input compares to the reference voltage.
- For this example, reference is 2.5 V, output is at op amp saturation at ±12 V




$$v_{out} = \begin{cases} 12V, & v_{in} < 2.5V \\ -12V, & v_{in} > 2.5V \end{cases}$$

# Comparator



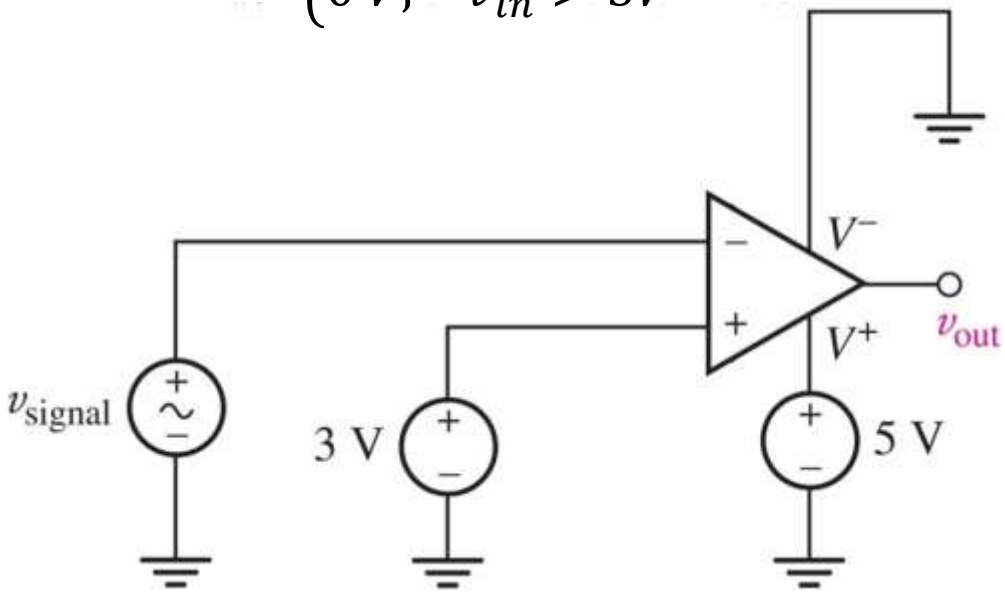

Very narrow, almost consider it as a vertical line

# Example 4

- Design a circuit that provides a “logic 1” 5 V output if a certain voltage signal drops below 3 V, and zero volts otherwise.

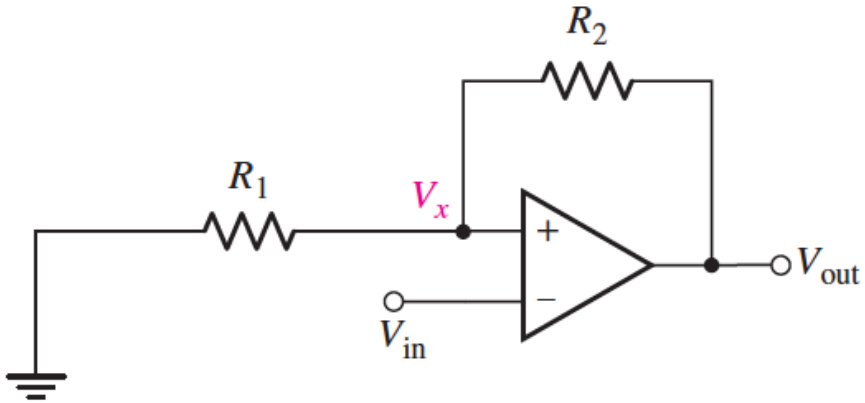
Solution:

$$v_{out} = \begin{cases} 5\ V, & v_{in} < 3V \\ 0\ V, & v_{in} > 3V \end{cases}$$



# Schmitt Trigger

- Comparator circuits can also be used in a positive feedback configuration, such as Schmitt trigger circuit:

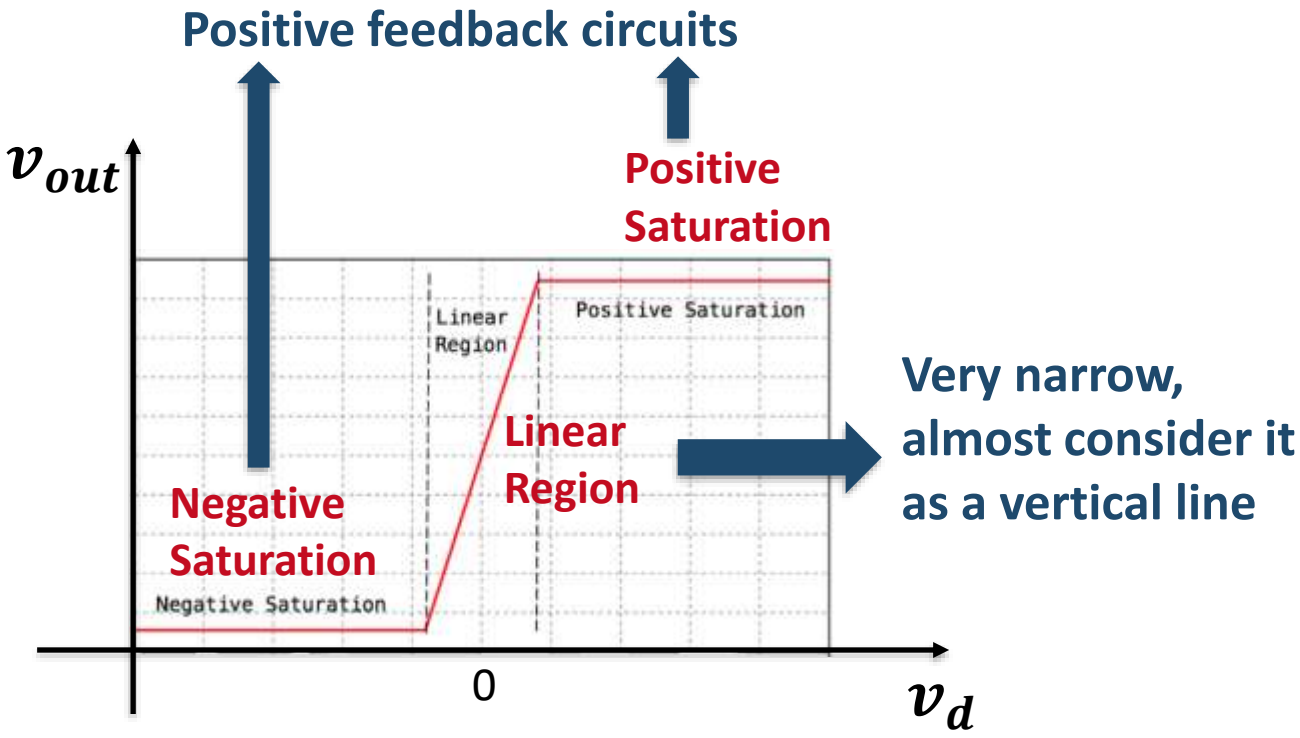



- The output voltage will be forced to the supply voltage $V^+$ and $V^-$,depending on the sign of the voltage difference between $V_x$ and $V_{in}$.

$$v_{out} = \begin{cases} V^-, & v_{in} > V_x \\ V^+, & v_{in} < V_x \end{cases} \qquad [1]$$



# Feedback circuit and saturation



# Schmitt Trigger

- The voltage at the non-inverting terminal $V_x$ is:

$$V_x = \frac{R_1}{R_1+R_2} V_{out} \qquad [2]$$

- Assuming that $V_{out} = V^+$ at initial state, the output voltage will remain in high state until $V_{in}$ exceeds $V_x$ . The upper threshold voltage is given by:

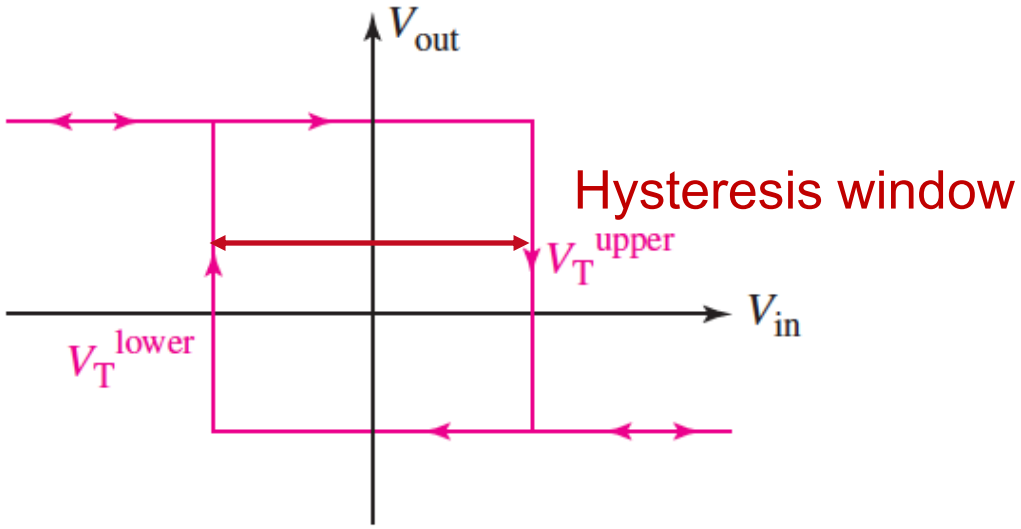
$$V_{in} = V_T^{upper} = \frac{R_1}{R_1 + R_2} V^+ \qquad [3]$$

- According to Eq [1], when $V_{in}$ exceeds $V_x$ , $V_{out}$ will be forced to be $V^-$. The output voltage will remain in low state until $V_{in}$ is less than $V_x$ . The lower threshold voltage is given by:

$$V_{in} = V_T^{lower} = \frac{R_1}{R_1 + R_2} V^- \qquad [4]$$



# Schmitt Trigger




- Circuit contains two thresholds for $V_x$: $V_T^{lower}$, $V_T^{upper}$.
- A comparator with positive feedback can lead to hysteresis (has "memory"). It reduces the sensitivity to fluctuations of input signal.



# THANK YOU

Xi'an Jiaotong-Liverpool University
西交利物浦大學